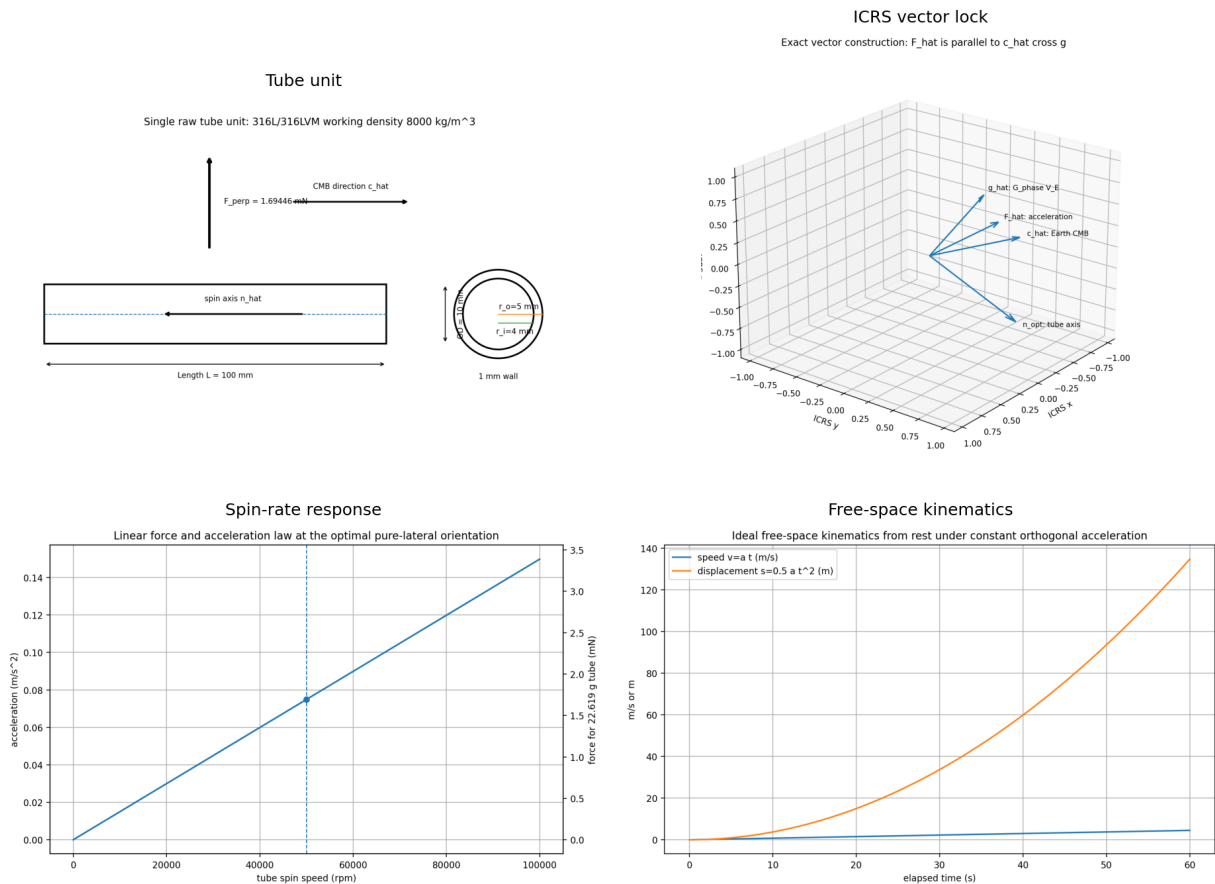


K-FIELD

Tubular Rotor Free-Space Forces

Single-unit mass, spin inertia, optimal Earth-CMB-orthogonal acceleration, and characteristic scaling laws



Title figure. Every panel is generated from the exact tube dimensions, calibrated Earth-CMB field inputs, and the computed free-space force and acceleration ledgers.

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This paper is a standalone educational and archival reference. It defines the exact Cell3 geometry and field inputs used by the force law, the dimensions and mass of a single surgical-stainless tubular rotor, the optimal pure-lateral orientation relative to the Earth CMB traversal vector, the resulting force and acceleration vectors, the raw tube spin inertia, angular momentum, stored spin energy, free-space motion laws, geometry laws, and the transformation from a free translating tube to a constrained torque-producing blade. All numerical figures and tables are generated from the numerical ledger contained in the machine-readable companion file.

Abstract

A thin-walled tubular rotor contains every circumferential phase simultaneously. Integrating the elemental force law around the complete circular wall removes the rotating pulse and leaves a steady lateral force proportional to total tube mass and signed axial angular velocity. For a free tube, division by mass removes the material mass from the acceleration law. The tube mass sets the absolute force, while the free acceleration is controlled by field orientation and spin rate.

The worked unit has 10 mm outside diameter, 8 mm inside diameter, 1 mm wall thickness, 100 mm length, and a working surgical-stainless density of 8000 kg/m³. Its raw steel mass is 22.619467 g. At 50,000 rpm, the tube angular speed is 5235.987756 rad/s. In the optimal pure-lateral orientation, the calculated steady force is 1.694456 mN and the free acceleration is 0.074911384 m/s², or 7.63884 milli-g. The acceleration vector is exactly orthogonal to the Earth CMB traversal vector to numerical precision.

The axial spin moment of inertia is 4.636990757e-7 kg m², the stored spin energy is 6.356286719 J, and the axial angular momentum is 2.427922683e-3 kg m²/s. Spin inertia determines the energy, angular momentum, and drive torque required to establish the rotation. It does not appear independently in the lateral acceleration law; the relevant ratio H/I is the signed angular velocity.

Central result

For a fixed field orientation and signed axial spin rate, the raw tube force is proportional to tube mass, while the free orthogonal acceleration is independent of tube mass: $F = m a$ and $a = (\kappa_F \omega_s / 2) \hat{n} \text{ cross } (G_{\text{phase}} V_E)$. Reversing tube spin reverses the complete acceleration vector by 180 degrees.

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1. Scope, definitions, and coordinate lock

The subject is one complete thin-walled tube rotating about its own longitudinal axis. The tube is considered first as a free body in a frictionless environment. It has no hub lever arm and no external translational constraint. The complete cross-section is integrated, so the output is a steady center-of-mass force rather than a rotating point-force pulse.

The coordinate frame is ICRS/J2000 Cartesian. The Earth CMB traversal direction is represented by the unit vector $\mathbf{c_hat}$. The Cell3 response matrix $\mathbf{G_phase}$ acts on the Earth velocity vector $\mathbf{V_E}$ to form the field-response vector $\mathbf{g}$.

$$\begin{aligned} \mathbf{g} &= \mathbf{G_phase} \mathbf{V_E} \\ \mathbf{F_tube} &= (\kappa_F m \omega_s / 2) \mathbf{n_hat} \text{ cross } \mathbf{g} \\ \mathbf{a_tube} &= \mathbf{F_tube} / m = (\kappa_F \omega_s / 2) \mathbf{n_hat} \text{ cross } \mathbf{g} \end{aligned}$$

The sign of ω_s carries the tube spin direction. A positive value refers to the adopted right-hand convention about $\mathbf{n_hat}$. Reversing the physical spin changes ω_s to $-\omega_s$ and reverses force and acceleration without changing magnitude.

Coordinate and field inputs

Quantity	Value	Unit or role
Earth CMB unit vector $\mathbf{c_hat}$	(-0.970770855836, 0.207348009111, -0.120874929482)	unitless ICRS
Earth CMB RA, Dec	167.943299880 deg, -6.942599922 deg	direction
Earth velocity $\mathbf{V_E}$	(-359003.199066, 76679.886034, -44701.060101)	m/s ICRS
Field response $\mathbf{g}=\mathbf{G_phase} \mathbf{V_E}$	(-880221.981635, -143911.436562, 266335.026949)	m/s effective response
Magnitude $ \mathbf{g} $	930825.217270434834	m/s effective response
Recommended κ_F	3.0740508997020345e-11	calibrated coupling
Calibration interval	[3.0427399447546500e-11, 3.1053618546494197e-11]	coupling interval

Exact vector construction: $\hat{F}$ is parallel to $\hat{c}$ cross $\hat{g}$

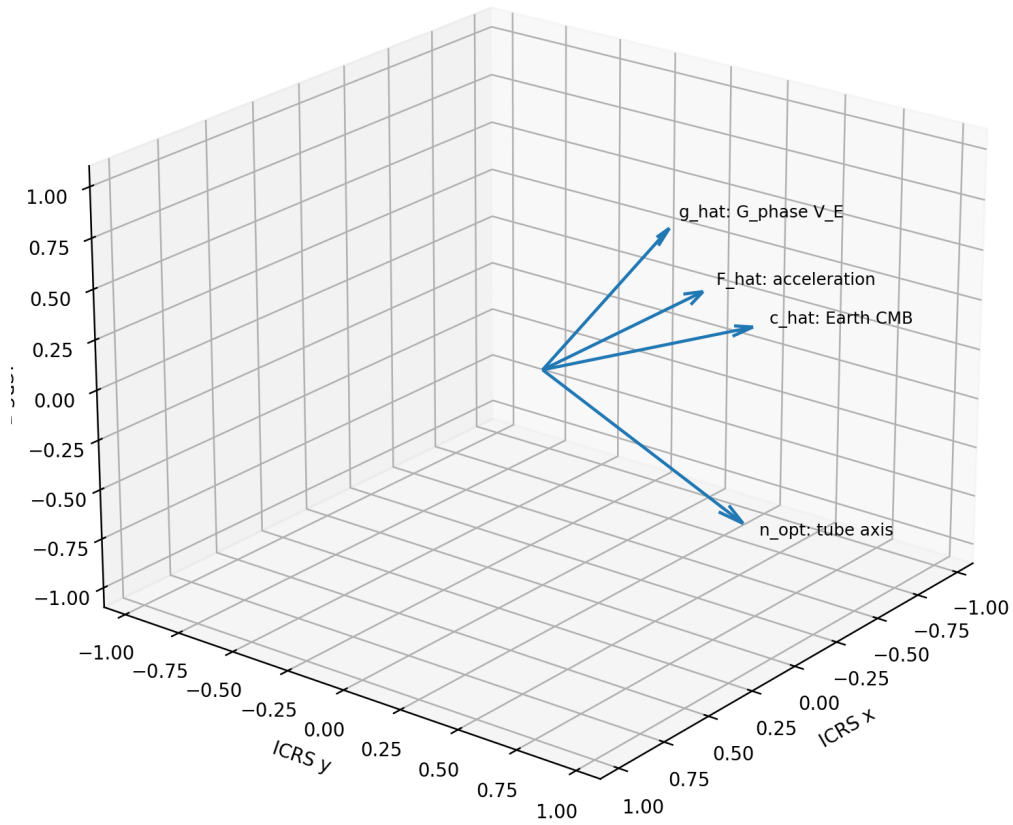


Figure 1. Exact ICRS vector geometry. The optimal tube axis $\hat{n}_{\text{opt}}$ is perpendicular to $\hat{g}$. The force and free acceleration are parallel to $\hat{c}_{\text{hat}}$ cross $\hat{g}$ and are therefore perpendicular to both the Earth CMB direction and $\hat{g}$.

2. Native Cell3 geometry and field planes

Cell3 supplies the anisotropic three-vector geometry associated with the physical response matrix. The centered base cell is generated from the basis matrix B whose columns are a , b , and c . Every address $L=(L_1,L_2,L_3)$, with each index in $\{-1,0,+1\}$, maps to the centered coordinate $p_L=B L/2$. The resulting 27 nodes consist of eight corners, twelve edge centers, six face centers, and one body center.

```
p_L = B L / 2, L_i in {-1,0,+1}
G_direct = B^T B
cell volume = |det(B)|
```

Direct basis vectors

Vector	ICRS x	ICRS y	ICRS z	Length
a	0.020381438516946	-0.003416414496411	-0.002332965916729	0.020797058782433
b	0.002198745968323	0.012902170258693	-0.034066501536056	0.036494205130727
c	-0.000630379485347	-0.024257722017128	-0.042924354571094	0.049308565899859

Primitive volume: 2.842066937809649e-05 native³.

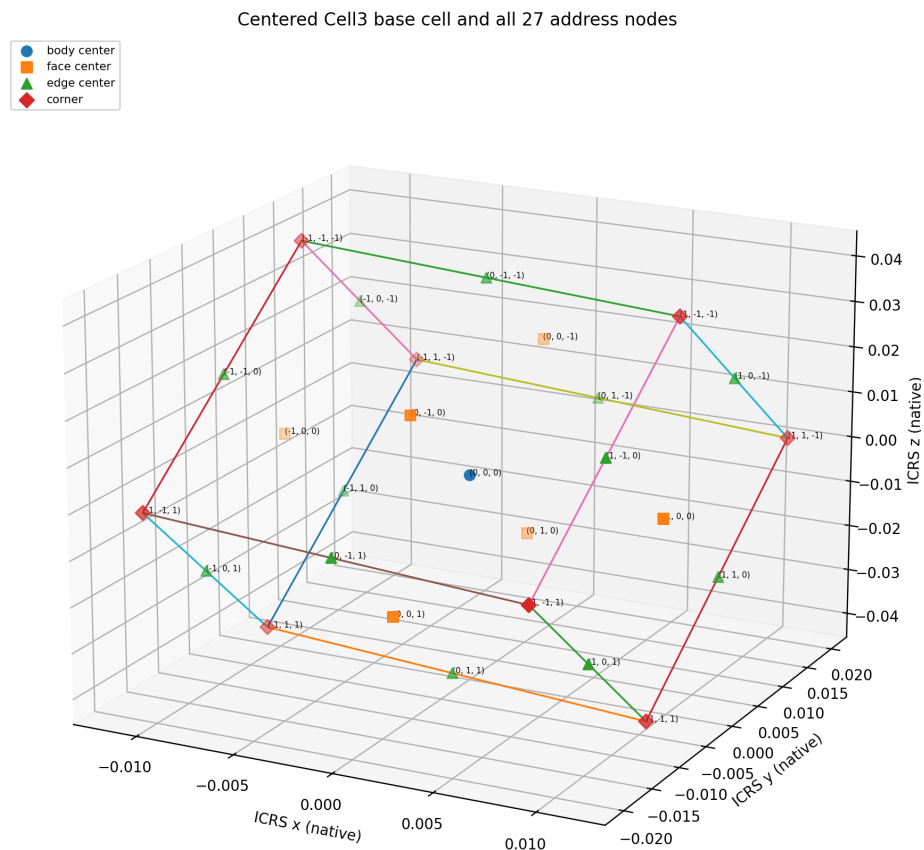


Figure 2. Dimensionally correct centered Cell3 base cell in the fixed ICRS frame. All 27 address nodes are plotted and labeled by integer index.

Primitive face geometry

Face	Included angle	Area	Unit normal (ICRS)
ab	83.933483012 deg	$7.547217896726240 \times 10^{-4} \text{ native}^2$	(0.194092200808, 0.913177168821, 0.358379234791)
ac	80.448130018 deg	$1.011255757688065 \times 10^{-3} \text{ native}^2$	(0.089052594102, 0.866576769291, -0.491033948326)
bc	50.363231814 deg	$1.385784431733287 \times 10^{-3} \text{ native}^2$	(-0.995965190309, 0.083602162507, -0.032619290564)

Dimensionally correct primitive Cell3 face planes

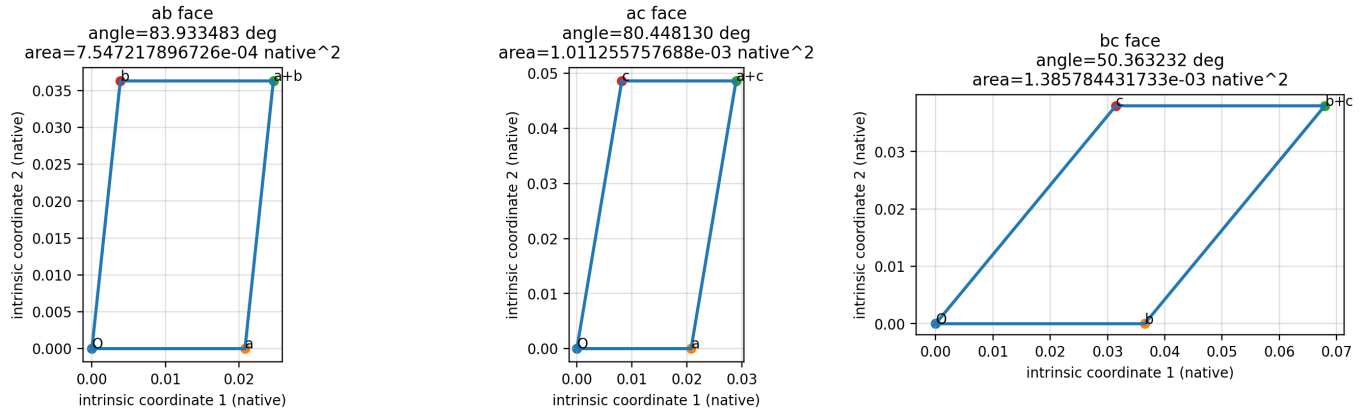


Figure 3. Dimensionally correct primitive face planes. Each parallelogram is constructed from the exact edge lengths and included angle of its basis-vector pair.

Physical response matrix

row	column 1	column 2	column 3
1	2.9764106378	1.2431507795	-2.0803600354
2	1.2431507795	2.5922286635	-2.3178839937
3	-2.0803600354	-2.3178839937	6.7735709133

3. Raw tubular rotor construction

The raw tube is the steel wall only. The calculation excludes end caps, shaft extensions, bearings, couplers, motor rotor mass, instrumentation, and support hardware. The working material density is 8000 kg/m³.

```

r_o = D_o / 2
r_i = r_o - t
A_metal = pi (r_o^2 - r_i^2)
V_tube = A_metal L
m_tube = rho V_tube

```

Construction quantity	Value	Unit
Outside diameter D_o	0.010000	m
Wall thickness t	0.001000	m
Inside diameter D_i	0.008000	m
Length L	0.100000	m
Density rho	8000.000000	kg/m ³
Metal cross-sectional area	2.827433388230814e-05	m ²
Raw steel volume	2.827433388230815e-06	m ³
Raw steel mass	2.261946710584652e-02	kg
Raw steel mass	22.619467106	g

Single raw tube unit: 316L/316LVM working density 8000 kg/m³

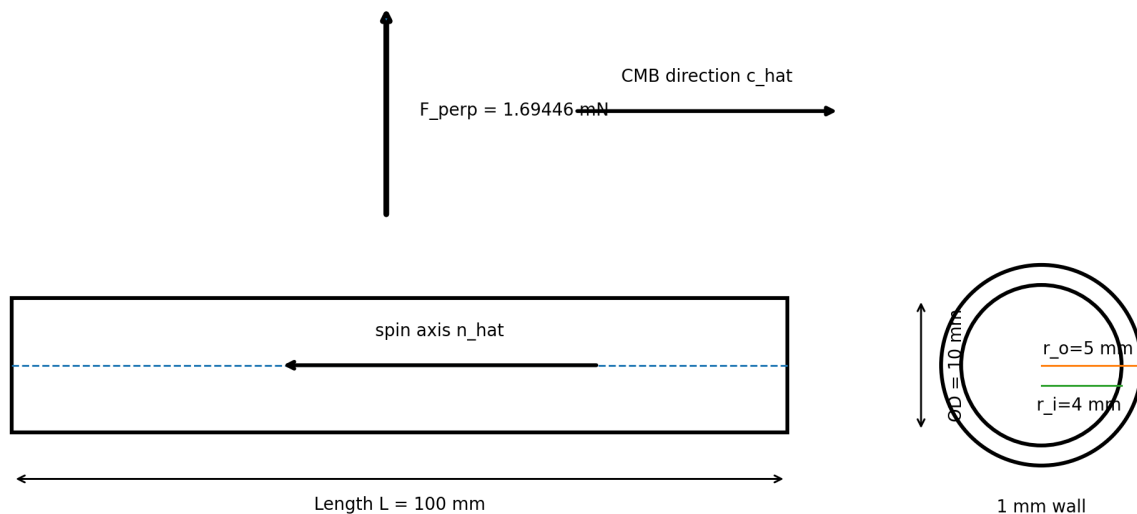


Figure 4. Dimensionally correct raw tube geometry and force direction. Dimensions are the stated 10 mm OD, 8 mm ID, 1 mm wall, and 100 mm length.

4. Integrated force law for one complete tube

At every cross-section, the circular wall contains all azimuthal phases. Opposite mass elements reverse both radial and tangential unit vectors. The two sign reversals cancel inside the cross-term, so opposite elements contribute force in the same integrated direction. The oscillating terms cancel around the full circumference, while the cycle-average lateral vector remains.

$$dF = -\kappa_F \omega_s [t(\theta)^T G_{\text{phase}} V_E] u(\theta) dm$$

$$F_{\text{tube}} = (\kappa_F m_{\text{tube}} \omega_s / 2) \hat{n} \text{ cross } (G_{\text{phase}} V_E)$$

Axial integration contributes only the total tube mass. At fixed angular speed and orientation, length, density, wall thickness, and diameter affect the total force only through m_{tube} . The tube axis passes through its center of mass, and the integrated force produces no intrinsic torque about the tube center in the centered model.

Specific-force coefficient

$$q_g = \kappa_F |g| / 2$$

$$F_{\text{max}} = m_{\text{tube}} q_g \omega_s$$

$$a_{\text{max}} = q_g \omega_s$$

The numerical coefficient is $q_g = 1.430702048307761 \times 10^{-5} \text{ N/[kg (rad/s)]}$.

5. Optimal pure-lateral orientation

The objective is to make the force exactly orthogonal to the Earth CMB vector and simultaneously maximize its magnitude. The tube-axis unit vector must be perpendicular to g to attain the full cross-product magnitude. It must also lie in the plane spanned by $\hat{c}$ and g so that $\hat{n} \text{ cross } g$ is parallel to $\hat{c} \text{ cross } g$.

$$g_{\text{hat}} = g / |g|$$

$$n_{\text{opt}} = [\hat{c} - (\hat{c} \text{ dot } g_{\text{hat}}) g_{\text{hat}}] / \sqrt{1 - (\hat{c} \text{ dot } g_{\text{hat}})^2}$$

$$F_{\text{hat}} = (\hat{c} \text{ cross } g) / |\hat{c} \text{ cross } g|$$

This construction is the normalized projection of the Earth CMB vector into the plane perpendicular to g . It satisfies $n_{\text{opt}} \text{ dot } g = 0$ and forces the response direction to be perpendicular to the Earth CMB vector.

Optimal vectors

Vector	ICRS components	RA	Dec
n_{opt}	(-0.315865036407, 0.646162683024, -0.694768354089)	116.050867377 deg	-44.008760899 deg
F_{hat}	(0.077469585422, 0.747375863362, 0.659869519070)	84.082114314 deg	41.289922334 deg

Numerical orthogonality checks: $F_{\text{hat}} \text{ dot } \hat{c} = 1.467 \times 10^{-18}$; $n_{\text{opt}} \text{ dot } g = 4.121 \times 10^{-10}$ in the numerical vector units.

Orientation envelope

For a general tube-axis angle α relative to g , the response magnitude is proportional to $|\sin(\alpha)|$. Maximum response occurs at $\alpha = 90$ deg; the response vanishes when the tube axis is parallel or antiparallel to g .

$$|F(\alpha)| = m_{\text{tube}} q_g |\omega_s| |\sin(\alpha)|$$

$$|a(\alpha)| = q_g |\omega_s| |\sin(\alpha)|$$

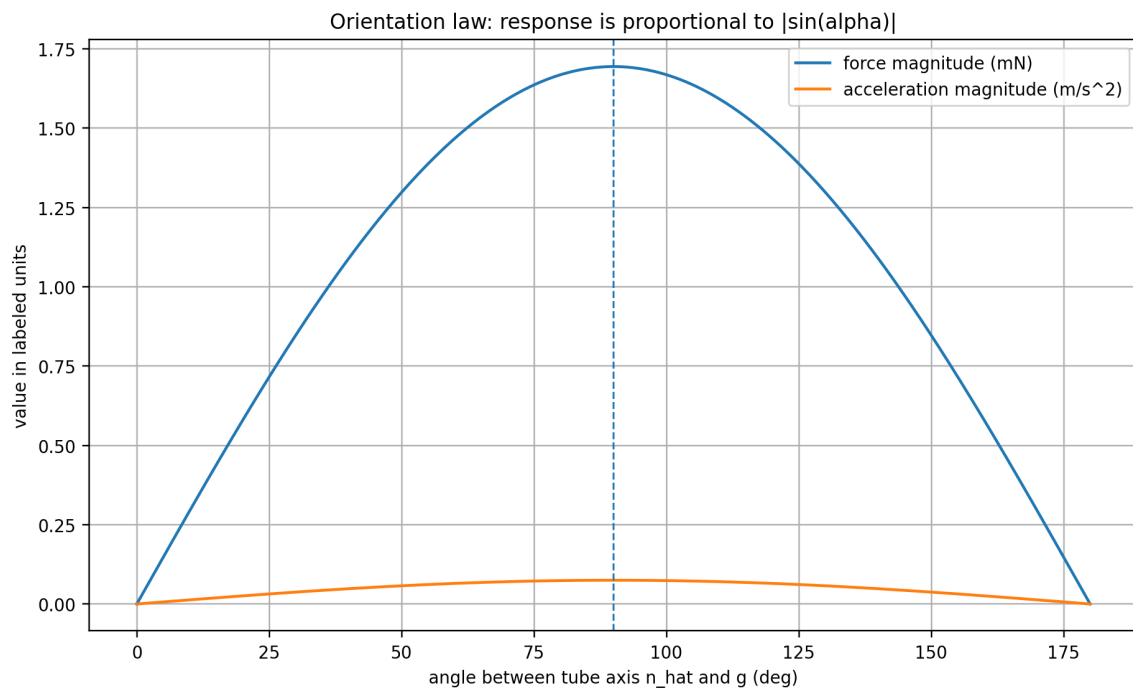


Figure 5. Force and acceleration orientation envelope for the defined tube at 50,000 rpm. The full response occurs when $\hat{n}$ is perpendicular to g .

6. Calibrated force vector

At 50,000 rpm, the signed angular speed is 5235.987756 rad/s. In the optimal orientation the cross-product magnitude equals $|g|$, so the tube develops the maximum force available to its mass and spin rate.

$$F_{\max} = (\kappa_F m_{\text{tube}} \omega_s / 2) |g|$$

$$F_{\text{vector}} = F_{\max} \hat{F}$$

Force quantity	Value	Unit
Nominal force magnitude	1.694455587915101e-03	N
Nominal force magnitude	1.694455588	mN
Calibration lower bound	1.677196595	mN
Calibration upper bound	1.711714581	mN
ICRS x component	1.312687719109319e-04	N
ICRS y component	1.266395207947365e-03	N
ICRS z component	1.118119593883255e-03	N
Force RA	84.082114314	deg
Force Dec	41.289922334	deg

The 22.619 g raw tube produces a steady 1.69446 mN lateral force at 50,000 rpm. The force direction is fixed in ICRS while the tube continues spinning about its own axis.

7. Free translational acceleration

For an unconstrained tube, Newtonian center-of-mass kinematics use $a=F/m$. Because the integrated K-Field force is proportional to total tube mass, mass cancels exactly.

$$a_{\text{vector}} = F_{\text{vector}} / m_{\text{tube}}$$

$$a_{\text{vector}} = (\kappa_F \omega_s / 2) n_{\text{opt}} \text{ cross } g$$

$$a_{\max} = q_g \omega_s$$

Acceleration quantity	Value	Unit
Magnitude	0.074911384073992	m/s ²
Magnitude	7.638835288	milli-g
Calibration lower bound	0.074148369066006	m/s ²
Calibration upper bound	0.075674399081979	m/s ²
ICRS x component	0.005803353867563	m/s ²
ICRS y component	0.055986960347976	m/s ²
ICRS z component	0.049431738981784	m/s ²

The mass law is therefore F proportional to m , while a is independent of m at fixed spin and orientation. A longer or denser tube develops proportionally greater force but undergoes the same free acceleration.

Mass increases absolute force and inertial resistance in the same proportion. It does not change the free acceleration of a geometrically complete tube operated at the same signed angular speed and orientation.

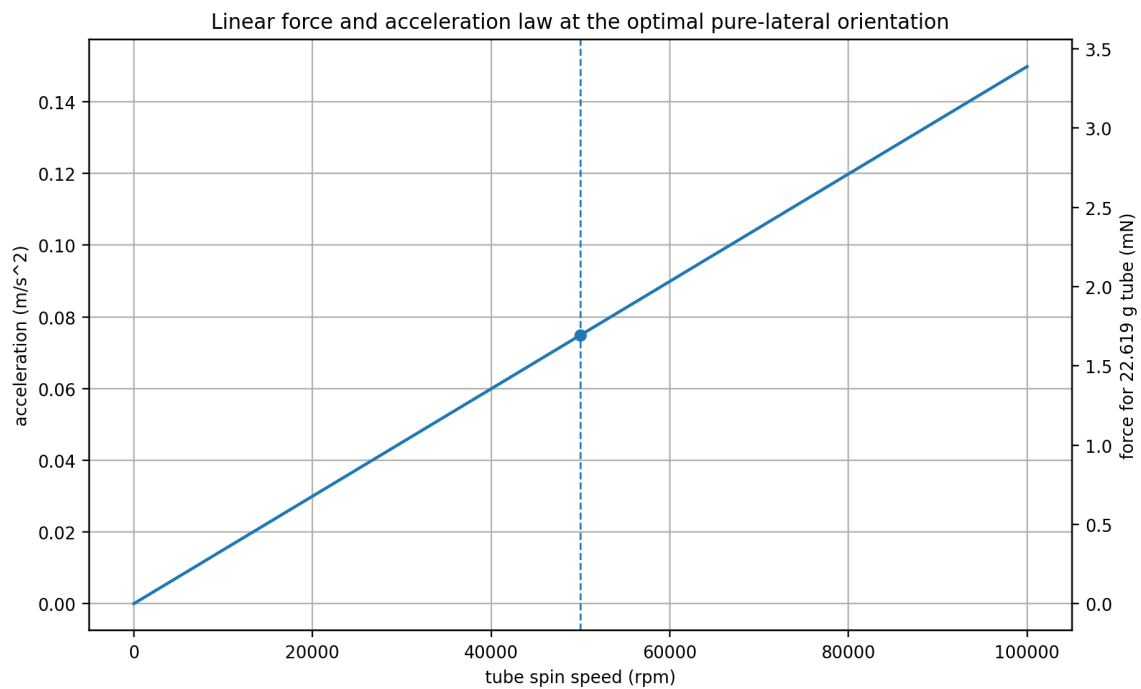


Figure 6. Linear spin-rate law. Acceleration is a property of spin rate and orientation; force is the same specific acceleration multiplied by the 22.619 g tube mass.

8. Free-space kinematics

If the tube begins from rest, its orientation remains locked, and the field inputs remain constant over the interval, the acceleration vector is constant. The standard constant-acceleration equations apply component by component.

$$v(t) = v_0 + a t$$

$$x(t) = x_0 + v_0 t + (1/2) a t^2$$

For release from rest, the scalar speed along F_{hat} is $v(t)=0.0749113841 t$ and the displacement is $s(t)=0.0374556920 t^2$.

Worked free-flight values

Elapsed time	Speed	Displacement	ICRS displacement vector
0.1 s	0.007491138 m/s	0.000374557 m	(0.000029, 0.000280, 0.000247) m
1 s	0.074911384 m/s	0.037455692 m	(0.002902, 0.027993, 0.024716) m
5 s	0.374556920 m/s	0.936392301 m	(0.072542, 0.699837, 0.617897) m
10 s	0.749113841 m/s	3.745569204 m	(0.290168, 2.799348, 2.471587) m
30 s	2.247341522 m/s	33.710122833 m	(2.611509, 25.194132, 22.244283) m
60 s	4.494683044 m/s	134.840491333 m	(10.446037, 100.776529, 88.977130) m

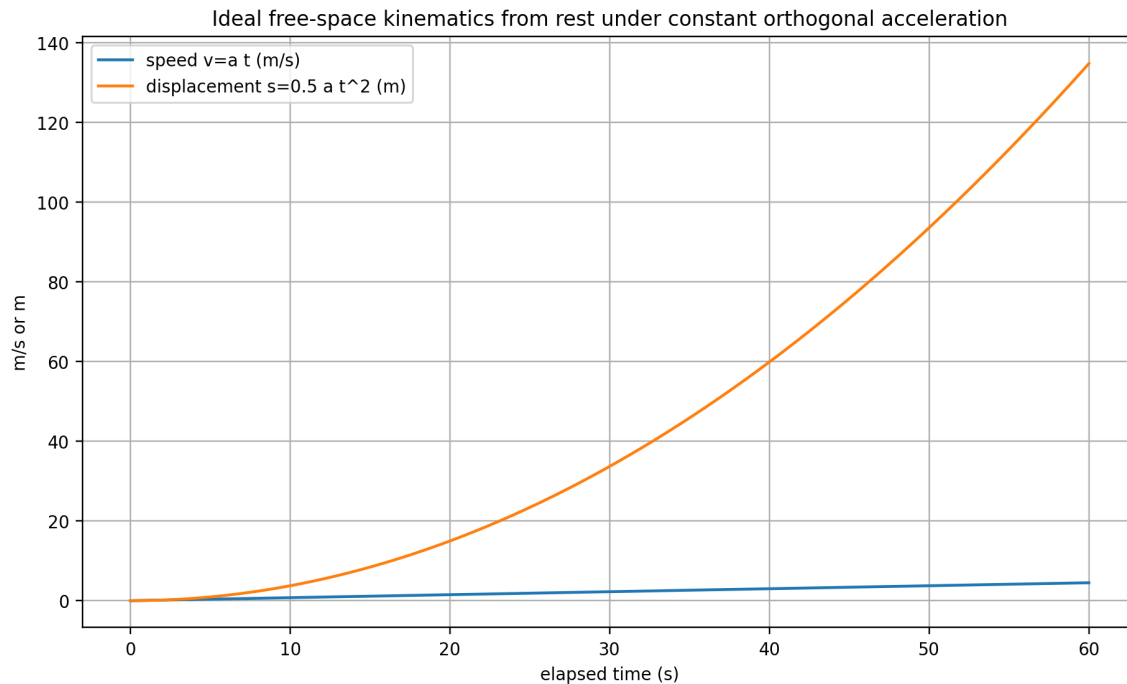


Figure 7. Ideal free-space speed and displacement from rest. Speed grows linearly; displacement grows quadratically.

9. Axial spin inertia

The raw tube spin inertia about its longitudinal axis is the hollow-cylinder moment of inertia. This inertia controls the torque required to change the tube spin rate.

$$I_s = (1/2) m_{\text{tube}} (r_o^2 + r_i^2)$$

$$I_s / m_{\text{tube}} = (1/2)(r_o^2 + r_i^2)$$

Inertia quantity	Value	Unit
Axial moment of inertia I_s	4.636990756698536e-07	kg m ²
Specific axial inertia I_s/m	2.050000000000000e-05	m ²

Inertia quantity	Value	Unit
Outer radius	0.005000000	m
Inner radius	0.004000000	m

Spin inertia does not appear as an independent multiplier in the free acceleration law. It enters through the relationship between stored angular momentum, stored energy, and angular velocity.

10. Spin angular momentum and stored energy

$H_s = I_s \omega_s$
 $E_s = (1/2) I_s \omega_s^2$

Spin-state quantity	Value	Unit
Spin speed	50000.000000	rpm
Spin frequency	833.333333333333	Hz
Spin period	1.200000000000e-03	s
Angular speed	5235.987755982988347	rad/s
Outer surface speed	26.179938779915	m/s
Inner surface speed	20.943951023932	m/s
Outer centripetal acceleration	1.370778389040e+05	m/s^2
Inner centripetal acceleration	1.096622711232e+05	m/s^2
Angular momentum magnitude H_s	2.427922682667983e-03	kg m^2/s
Specific angular momentum	0.107337748997651	m^2/s
Stored spin energy E_s	6.356286719461465	J
Specific spin energy	281.009569753238679	J/kg

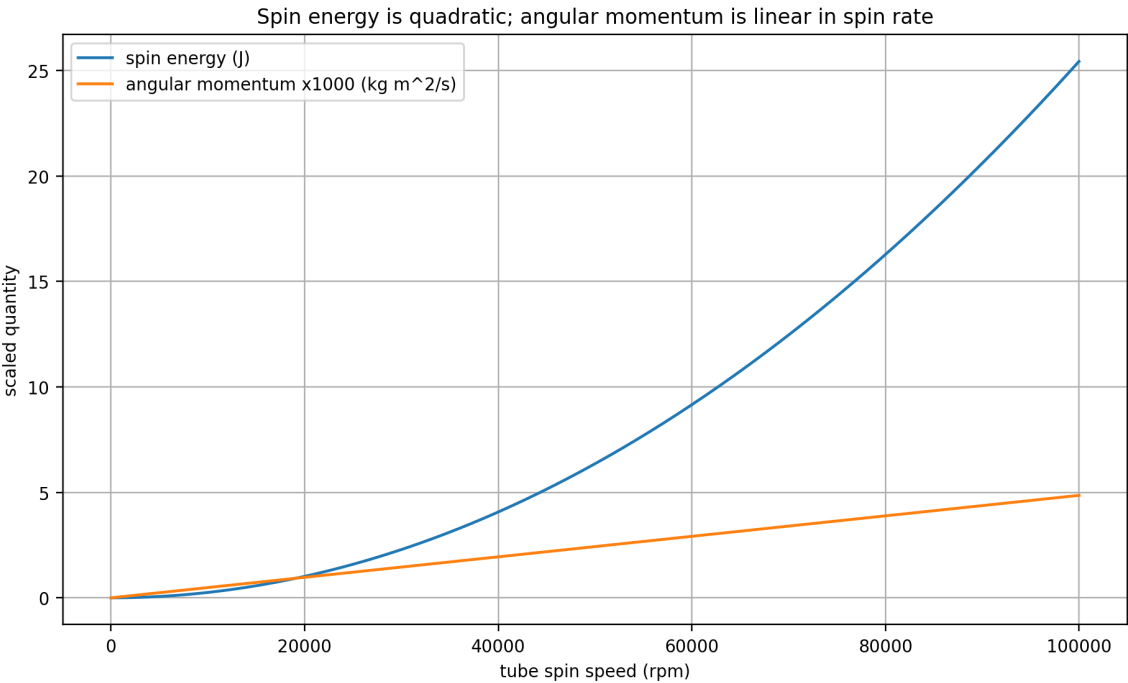


Figure 8. Angular momentum rises linearly with RPM, while stored rotational energy rises quadratically.

11. Acceleration versus spin rate

At the optimal orientation, acceleration is exactly linear in signed axial angular speed.

```
a_perp = q_g omega_s
omega_s = (2 pi / 60) RPM
a_perp = 1.49822768148e-6 RPM m/s^2
```

Spin speed	Angular speed	Acceleration	Force
0 rpm	0.000000 rad/s	0.000000000 m/s^2	0.000000000 mN
10,000 rpm	1047.197551 rad/s	0.014982277 m/s^2	0.338891118 mN
25,000 rpm	2617.993878 rad/s	0.037455692 m/s^2	0.847227794 mN
50,000 rpm	5235.987756 rad/s	0.074911384 m/s^2	1.694455588 mN
75,000 rpm	7853.981634 rad/s	0.112367076 m/s^2	2.541683382 mN
100,000 rpm	10471.975512 rad/s	0.149822768 m/s^2	3.388911176 mN

The sign of acceleration follows the sign of RPM. Doubling RPM doubles force and acceleration, doubles angular momentum, and quadruples stored spin energy.

12. Acceleration versus tube surface velocity

```
v_o = r_o omega_s
a_perp = q_g v_o / r_o
```

For a fixed tube radius, acceleration is linear in outer surface speed. For a fixed permitted surface speed, reducing the tube radius increases angular speed and acceleration as $1/r_o$. This is the narrow-tube scaling. It is distinct from mass scaling because the free acceleration remains independent of total tube mass.

For the defined tube, $v_o=26.179938780$ m/s and $a_{\text{perp}}=0.074911384$ m/s².

13. Acceleration versus spin energy

Eliminating angular speed through the spin-energy equation gives a direct relationship between acceleration and stored rotational energy.

```
omega_s = sqrt(2 E_s / I_s)
a_perp = q_g sqrt(2 E_s / I_s)
a_perp = 2 q_g sqrt[e_s / (r_o^2+r_i^2)] where e_s=E_s/m_tube
```

At fixed tube geometry, acceleration grows with the square root of specific spin energy. If total stored energy rather than specific energy is held fixed across otherwise identical tubes, I_s is proportional to mass, ω_s is proportional to $1/\sqrt{m_{\text{tube}}}$, and therefore a_{perp} is proportional to $1/\sqrt{m_{\text{tube}}}$ while F_{perp} is proportional to $\sqrt{m_{\text{tube}}}$. Because spin energy is quadratic in angular speed while force is linear, force per stored joule decreases as spin speed rises.

```
fixed total E_s and fixed geometry: a_perp proportional to 1/sqrt(m_tube)
fixed total E_s and fixed geometry: F_perp proportional to sqrt(m_tube)
```

14. Acceleration versus angular momentum

```
omega_s = H_s / I_s
a_perp = q_g H_s / I_s
a_perp = 2 q_g h_s / (r_o^2+r_i^2) where h_s=H_s/m_tube
```

The ratio H_s/I_s is simply the signed angular velocity. A larger moment of inertia does not create greater acceleration by itself. For a fixed H_s , increasing I_s lowers angular speed and therefore lowers acceleration.

15. Mass and geometry laws

$$\begin{aligned} m_{\text{tube}} &= \rho \pi (r_o^2 - r_i^2) L \\ F_{\text{perp}} &= q_g \omega_s \rho \pi (r_o^2 - r_i^2) L \\ a_{\text{perp}} &= q_g \omega_s \end{aligned}$$

At fixed angular velocity, force is linear in density, tube length, and metal cross-sectional area. Free acceleration is independent of all three because force and inertial mass rise together.

For a thin wall with $r_i = r_o - t$:

$$\begin{aligned} r_o^2 - r_i^2 &= 2 r_o t - t^2 \\ m_{\text{tube}} &\text{approximately } 2 \pi \rho r_o t L \\ F_{\text{perp}} &\text{approximately } 2 \pi q_g \omega_s \rho r_o t L \end{aligned}$$

The thin-wall approximations become exact in the limit t/r_o approaches zero. The worked tube uses the exact annular formula rather than the approximation.

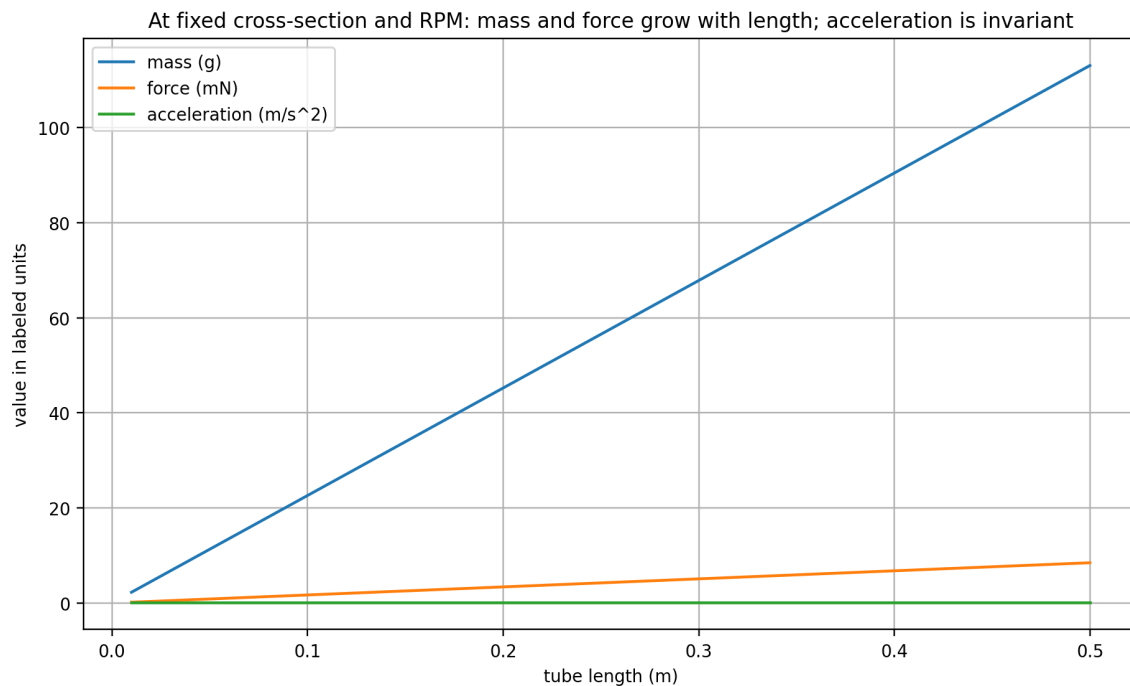


Figure 9. Tube-length sweep for the fixed 10 mm OD, 1 mm wall cross-section at 50,000 rpm. Mass, force, inertia, and stored energy grow linearly with length; free acceleration remains constant.

16. Force-inertia identities

The tube mass may be eliminated in favor of measurable axial inertia.

$$\begin{aligned} m_{\text{tube}} &= 2 I_s / (r_o^2 + r_i^2) \\ F_{\text{perp}} &= 2 q_g I_s \omega_s / (r_o^2 + r_i^2) \\ F_{\text{perp}} &= 2 q_g H_s / (r_o^2 + r_i^2) \end{aligned}$$

These identities are algebraic representations of the same mass-dependent force law. They do not assign causal primacy to inertia or angular momentum. The fundamental response remains linear in mass and signed angular velocity.

17. Force-to-energy characteristic

$$F_{\text{perp}} / E_s = 4 q_g / [(r_o^2 + r_i^2) \omega_s]$$

For the defined tube at 50,000 rpm, $F_{\text{perp}}/E_s = 2.665794767764446 \times 10^{-4}$ N/J.

At fixed geometry, force increases as ω_s while stored energy increases as ω_s^2 . Therefore force is proportional to $\sqrt{E_s}$, and force per stored joule is proportional to $1/\omega_s$.

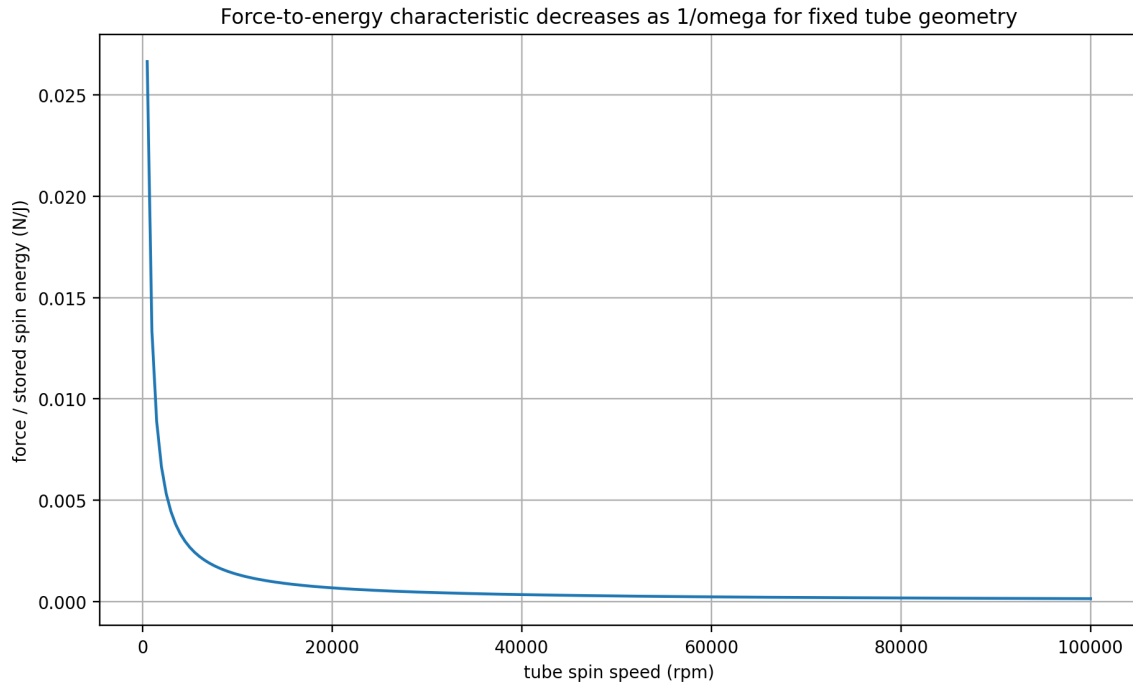


Figure 10. Force-to-stored-spin-energy ratio for the defined raw tube. The ratio diverges mathematically near zero RPM because stored energy is quadratic while force is linear; both force and useful absolute output approach zero there.

18. Spin-up and steady-state drive requirements

Spin inertia determines the torque and power required to establish the 50,000 rpm operating state. For an ideal constant angular acceleration from rest over spin-up time T_s :

$$\begin{aligned}\alpha_s &= \omega_s / T_s \\ \tau_{drive} &= I_s \alpha_s = I_s \omega_s / T_s \\ \text{average spin-up power} &= E_s / T_s \\ \text{final instantaneous spin-up power} &= \tau_{drive} \omega_s = 2 E_s / T_s\end{aligned}$$

Spin-up time T_s	Required ideal torque	Average ideal power	Final instantaneous power
0.1 s	2.427922683e-02 N m	63.562867195 W	127.125734389 W
1 s	2.427922683e-03 N m	6.356286719 W	12.712573439 W
5 s	4.855845365e-04 N m	1.271257344 W	2.542514688 W
10 s	2.427922683e-04 N m	0.635628672 W	1.271257344 W
60 s	4.046537804e-05 N m	0.105938112 W	0.211876224 W

After the target spin rate is reached, an ideal frictionless rotor requires no continuing drive power merely to preserve axial rotation. Real operation requires replacement of bearing, windage, eddy-current, and electrical losses. The lateral center-of-mass force is not a braking torque about the tube spin axis in the centered integrated model.

19. Mounted-blade transformation

The same tube may be constrained at a hub lever arm r_h . The support prevents the free center-of-mass translation and transfers the lateral force into a moment about the hub.

$$\begin{aligned}\tau_h &= r_h F_{\text{perp}} \text{ when } F_{\text{perp}} \text{ is tangential to the hub path} \\ P_h &= \tau_h \omega_h = F_{\text{perp}} v_t\end{aligned}$$

For the present tube, $\tau_h = 1.694455587915101e-03 r_h$ N m with r_h in metres. The tube force remains 1.694455588 mN; changing the lever arm changes torque, not the underlying single-unit lateral force.

The free-body and mounted-body descriptions are therefore two uses of one interaction:

$$\begin{aligned}\text{free tube: } F &= m a \\ \text{mounted tube: } \tau &= r \text{ cross } F\end{aligned}$$

20. Reversal, symmetry, and vector sign

The integrated force is odd in signed angular speed.

$$\begin{aligned}\omega_s &\rightarrow -\omega_s \\ F &\rightarrow -F \\ a &\rightarrow -a\end{aligned}$$

The forward acceleration direction is RA=84.082114314 deg, Dec=41.289922334 deg. Reversing spin changes the direction to RA=264.082114314 deg, Dec=-41.289922334 deg.

The magnitude, tube mass, spin inertia, stored energy, and absolute angular momentum are unchanged. Only the signed angular momentum, force vector, and acceleration vector reverse.

21. Measurement and experimental observables

The single-unit experiment separates the translational response from hub mechanics. The tube should be operated at the optimal axis orientation, then tested under spin reversal and RPM sweeps. The defining signatures are linear acceleration versus signed RPM, force proportional to raw tube mass, mass-independent free acceleration, and an exact 180-degree vector reversal when spin reverses.

Primary observables

Observable	Predicted law	Defined-unit witness
Force versus RPM	$F = m \cdot q \cdot g \cdot (2 \pi / 60) \text{ RPM}$	1.694455588 mN at 50,000 rpm
Acceleration versus RPM	$a = q \cdot g \cdot (2 \pi / 60) \text{ RPM}$	0.074911384 m/s ² at 50,000 rpm
Force versus tube mass	F proportional to m	1.69446 mN for 22.619 g
Acceleration versus mass	a independent of m	0.074911384 m/s ²
Spin reversal	$a_{\text{reverse}} = -a_{\text{forward}}$	180 deg vector change
Free velocity	$v = a \cdot t$	0.749114 m/s after 10 s
Free displacement	$s = 0.5 \cdot a \cdot t^2$	3.745569 m after 10 s
Stored spin energy	$E = 0.5 \cdot I \cdot \omega^2$	6.356287 J
Angular momentum	$H = I \cdot \omega$	0.002427923 kg m ² /s

A complete measurement record should include tube dimensions and mass, spin RPM and sign, tube-axis ICRS orientation, laboratory orientation relative to c_{hat} and g, support constraint state, measured force or acceleration vector, temperature, bearing state, and time history.

22. Conclusions

The defined raw tube is a 22.619467 g hollow cylindrical rotor. At 50,000 rpm in the optimal pure-lateral orientation it develops a steady 1.694456 mN force in the fixed ICRS direction RA 84.082114 deg, Dec 41.289922 deg. The force is perpendicular to the Earth CMB traversal vector.

As a free body, the tube accelerates at 0.074911384 m/s². This acceleration does not depend on tube mass, density, length, or wall thickness at fixed signed angular velocity and orientation. Those properties scale force and inertial mass together.

The raw tube axial inertia is 4.636990757e-7 kg m². At operating speed it carries 2.427922683e-3 kg m²/s of axial angular momentum and stores 6.356286719 J of rotational energy. These quantities determine spin-up requirements and gyroscopic behavior. They do not independently alter the free lateral acceleration because their ratio reduces to angular velocity.

The characteristic laws are linear force and acceleration versus spin rate, quadratic stored energy versus spin rate, linear angular momentum versus spin rate, linear force versus raw tube mass, and exact 180-degree force reversal under spin reversal. When the same tube is mounted at a lever arm, its fixed lateral force becomes hub torque according to $\tau = r \cdot F$.

The complete single-unit result is: $m_{\text{tube}} = 0.0226194671 \text{ kg}$; $I_s = 4.636990757 \text{e-}7 \text{ kg m}^2$; $\omega_s = 5235.987756 \text{ rad/s}$; $F_{\text{perp}} = 1.694455588 \text{e-}3 \text{ N}$; $a_{\text{perp}} = 0.0749113841 \text{ m/s}^2$; $E_s = 6.356286719 \text{ J}$; $H_s = 2.427922683 \text{e-}3 \text{ kg m}^2/\text{s}$.

Appendix A. Complete numerical input ledger

A.1 Field inputs

Symbol	Value
kappa_F	3.0740508997020345e-11
kappa_F lower	3.0427399447546500e-11
kappa_F upper	3.1053618546494197e-11
g	9.3082521727043483e+05
q_g	1.4307020483077610e-05

A.2 G_phase matrix

row	1	2	3
1	2.9764106378	1.2431507795	-2.0803600354
2	1.2431507795	2.5922286635	-2.3178839937
3	-2.0803600354	-2.3178839937	6.7735709133

A.3 Vector ledger

Vector	x	y	z	RA	Dec
c_hat	-0.970770855836	0.207348009111	-0.120874929482	167.943299880	-6.942599922
V_E direction	-0.970770855836	0.207348009111	-0.120874929482	167.943299880	-6.942599922
g_hat	-0.945636157362	-0.154606293310	0.286127859460	189.285391176	16.626277960
n_opt	-0.315865036407	0.646162683024	-0.694768354089	116.050867377	-44.008760899
F_hat	0.077469585422	0.747375863362	0.659869519070	84.082114314	41.289922334

A.4 Tube and mechanical ledger

Quantity	Value	Unit
D_o	1.0000000000000000e-02	m
D_i	8.0000000000000002e-03	m
wall	1.0000000000000000e-03	m
L	1.0000000000000001e-01	m
rho	8.0000000000000000e+03	kg/m^3
mass	2.2619467105846516e-02	kg
I_spin	4.6369907566985360e-07	kg m^2
omega_s	5.2359877559829883e+03	rad/s
H_spin	2.4279226826679828e-03	kg m^2/s
E_spin	6.3562867194614645e+00	J
force	1.6944555879151008e-03	N
acceleration	7.4911384073992185e-02	m/s^2

Appendix B. All 27 Cell3 address nodes

Coordinates use the centered mapping $p_L=B L/2$ in native source units.

L1	L2	L3	ICRS x	ICRS y	ICRS z	Class	Lambda sum
-1	-1	-1	-0.010974902499961	0.007385983127423	0.039661911011939	corner	-3
-1	-1	0	-0.011290092242635	-0.004742877881141	0.018199733726392	edge center	-2
-1	-1	1	-0.011605281985309	-0.016871738889705	-0.003262443559155	corner	-1
-1	0	-1	-0.009875529515800	0.013837068256769	0.022628660243911	edge center	-2
-1	0	0	-0.010190719258473	0.001708207248205	0.001166482958364	face center	-1
-1	0	1	-0.010505909001147	-0.010420653760359	-0.020295694327182	edge center	0
-1	1	-1	-0.008776156531638	0.020288153386115	0.005595409475884	corner	-1
-1	1	0	-0.009091346274312	0.008159292377552	-0.015866767809663	edge center	0
-1	1	1	-0.009406536016985	-0.003969568631012	-0.037328945095210	corner	1
0	-1	-1	-0.000784183241488	0.00567775879218	0.038495428053575	edge center	-2
0	-1	0	-0.001099372984162	-0.006451085129346	0.017033250768028	face center	-1
0	-1	1	-0.001414562726835	-0.018579946137910	-0.004428926517519	edge center	0
0	0	-1	0.000315189742674	0.012128861008564	0.021462177285547	face center	-1
0	0	0	0.000000000000000	0.000000000000000	0.000000000000000	body center	0
0	0	1	-0.000315189742674	-0.012128861008564	-0.021462177285547	face center	1
0	1	-1	0.001414562726835	0.018579946137910	0.004428926517519	edge center	0
0	1	0	0.001099372984162	0.006451085129346	-0.017033250768028	face center	1
0	1	1	0.000784183241488	-0.00567775879218	-0.038495428053575	edge center	2
1	-1	-1	0.009406536016985	0.003969568631012	0.037328945095210	corner	-1
1	-1	0	0.009091346274312	-0.008159292377552	0.015866767809663	edge center	0
1	-1	1	0.008776156531638	-0.020288153386115	-0.005595409475884	corner	1
1	0	-1	0.010505909001147	0.010420653760359	0.020295694327182	edge center	0
1	0	0	0.010190719258473	-0.001708207248205	-0.001166482958364	face center	1
1	0	1	0.009875529515800	-0.013837068256769	-0.022628660243911	edge center	2
1	1	-1	0.011605281985309	0.016871738889705	0.003262443559155	corner	1
1	1	0	0.011290092242635	0.004742877881141	-0.018199733726392	edge center	2
1	1	1	0.010974902499961	-0.007385983127423	-0.039661911011939	corner	3

Appendix C. Spin-rate and time-domain ledgers

C.1 Selected spin-rate states

RPM	omega (rad/s)	Force (N)	Acceleration (m/s^2)	Energy (J)	H (kg m^2/s)
0	0.00000000	0.00000000000e+00	0.00000000000e+00	0.00000000000e+00	0.00000000000e+00
5000	523.598775598	1.694455587915e-04	7.491138407399e-03	6.356286719461e-02	2.427922682668e-04
10000	1047.197551197	3.388911175830e-04	1.498227681480e-02	2.542514687785e-01	4.855845365336e-04
15000	1570.796326795	5.083366763745e-04	2.247341522220e-02	5.720658047515e-01	7.283768048004e-04
20000	2094.395102393	6.777822351660e-04	2.996455362960e-02	1.017005875114e+00	9.711690730672e-04
25000	2617.993877991	8.472277939576e-04	3.745569203700e-02	1.589071679865e+00	1.213961341334e-03
30000	3141.592653590	1.016673352749e-03	4.494683044440e-02	2.288263219006e+00	1.456753609601e-03
35000	3665.191429188	1.186118911541e-03	5.243796885179e-02	3.114580492536e+00	1.699545877868e-03
40000	4188.790204786	1.355564470332e-03	5.992910725919e-02	4.068023500455e+00	1.942338146134e-03
45000	4712.388980385	1.525010029124e-03	6.742024566659e-02	5.148592242764e+00	2.185130414401e-03
50000	5235.987755983	1.694455587915e-03	7.491138407399e-02	6.356286719461e+00	2.427922682668e-03
55000	5759.586531581	1.863901146707e-03	8.240252248139e-02	7.691106930548e+00	2.670714950935e-03
60000	6283.185307180	2.033346705498e-03	8.989366088879e-02	9.153052876025e+00	2.913507219202e-03
65000	6806.784082778	2.202792264290e-03	9.738479929619e-02	1.074212455589e+01	3.156299487468e-03
70000	7330.382858376	2.372237823081e-03	1.048759377036e-01	1.245832197014e+01	3.399091755735e-03
75000	7853.981633974	2.541683381873e-03	1.123670761110e-01	1.430164511879e+01	3.641884024002e-03
80000	8377.580409573	2.711128940664e-03	1.198582145184e-01	1.627209400182e+01	3.884676292269e-03
85000	8901.179185171	2.880574499456e-03	1.273493529258e-01	1.836966861924e+01	4.127468560536e-03
90000	9424.777960769	3.050020058247e-03	1.348404913332e-01	2.059436897106e+01	4.370260828802e-03
95000	9948.376736368	3.219465617039e-03	1.423316297406e-01	2.294619505726e+01	4.613053097069e-03
100000	10471.975511966	3.388911175830e-03	1.498227681480e-01	2.542514687785e+01	4.855845365336e-03

C.2 Selected free-flight states from rest

Time (s)	Speed (m/s)	Displacement (m)	x (m)	y (m)	z (m)
0.0	0.00000000	0.000000000	0.000000000	0.000000000	0.000000000
5.0	0.374556920	0.936392301	0.072541923	0.699837004	0.617896737
10.0	0.749113841	3.745569204	0.290167693	2.799348017	2.471586949
15.0	1.123670761	8.427530708	0.652877310	6.298533039	5.561070635
20.0	1.498227681	14.982276815	1.160670774	11.197392070	9.886347796
25.0	1.872784602	23.409807523	1.813548084	17.495925109	15.447418432
30.0	2.247341522	33.710122833	2.611509240	25.194132157	22.244282542
35.0	2.621898443	45.883222745	3.554554244	34.292013213	30.276940126
40.0	2.996455363	59.929107259	4.642683094	44.789568278	39.545391185
45.0	3.371012283	75.847776375	5.875895791	56.686797352	50.049635719
50.0	3.745569204	93.639230092	7.254192334	69.983700435	61.789673727
55.0	4.120126124	113.303468412	8.777572725	84.680277526	74.765505210
60.0	4.494683044	134.840491333	10.446036962	100.776528626	88.977130167

Appendix D. Compact equation set and reconstruction procedure

Relationship	Equation
Tube mass	$m = \rho \pi (r_o^2 - r_i^2) L$
Spin inertia	$I = (1/2) m (r_o^2 + r_i^2)$
Angular speed	$\omega = (2 \pi / 60) \text{RPM}$
Angular momentum	$H = I \omega$
Spin energy	$E = (1/2) I \omega^2$
Field response	$\mathbf{g} = \mathbf{G_phase} V_E$
Tube force	$\mathbf{F} = (\kappa_F m \omega / 2) \mathbf{n} \times \mathbf{g}$
Free acceleration	$\mathbf{a} = (\kappa_F \omega / 2) \mathbf{n} \times \mathbf{g}$
Optimal axis	$\mathbf{n_opt} = [\mathbf{c} - (\mathbf{c} \cdot \mathbf{g_hat}) \mathbf{g_hat}] / \sqrt{1 - (\mathbf{c} \cdot \mathbf{g_hat})^2}$
Force direction	$\mathbf{F_hat} = (\mathbf{c} \times \mathbf{g}) / \mathbf{c} \times \mathbf{g} $
Maximum specific force	$q_g = \kappa_F g / 2$
Maximum force	$F_{\max} = m q_g \omega$
Maximum acceleration	$a_{\max} = q_g \omega$
Free velocity	$\mathbf{v} = \mathbf{v}_0 + \mathbf{a} t$
Free position	$\mathbf{x} = \mathbf{x}_0 + \mathbf{v}_0 t + (1/2) \mathbf{a} t^2$
Mounted torque	$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$
Mounted power	$P = \boldsymbol{\tau} \cdot \boldsymbol{\Omega} = \mathbf{F} \cdot \mathbf{v}_t$

Minimal reconstruction sequence

1. Load $\mathbf{G_phase}$, V_E , $\mathbf{c_hat}$, and κ_F .
2. Compute $\mathbf{g} = \mathbf{G_phase} V_E$ and $\mathbf{g_hat} = \mathbf{g} / |\mathbf{g}|$.
3. Compute $\mathbf{n_opt}$ as the normalized component of $\mathbf{c_hat}$ perpendicular to $\mathbf{g}$.
4. Compute $\mathbf{F_hat} = (\mathbf{c_hat} \times \mathbf{g}) / |\mathbf{c_hat} \times \mathbf{g}|$.
5. Compute tube mass and axial inertia from exact annular geometry.
6. Convert RPM to signed angular speed.
7. Compute force $m q_g \omega \mathbf{F_hat}$ and acceleration $q_g \omega \mathbf{F_hat}$.
8. Compute $H = I \omega$ and $E = (1/2) I \omega^2$.
9. Apply constant-acceleration kinematics for the selected free-flight interval.
10. Reverse ω to reverse the complete force and acceleration vectors.

Appendix E. Archival record

Field	Record
Document title	K-Field Tubular Rotor Free-Space Forces
Principal Investigator	P. Dwayne Esterline
Organization	Krytronx
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation UTC	2026-07-24T14:53:29Z
Generation epoch	1784904809
Output PDF	K-Field_tubular_rotor_free_space_forces_1784904809.pdf
Machine-readable companion	K-Field_tubular_rotor_free_space_forces_1784904809.json
Report generation	Python ReportLab; built-in core fonts only
Figure generation	Python numerical computation and raster rendering from the exact report ledger
Source format references	K-Field Tubular Rotor Force Analysis; K-Field Tubular Rotor Earth-CMB Windmill; prior archival K-Field report conventions
Coordinate frame	ICRS/J2000 Cartesian
Mechanical units	SI
Cell3 units	native source units

P. Dwayne Esterline

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